

CONCEPCION, Chile

WMO: 856820

Lat: 36.7783S

Lon: 73.0625W

Elev: 56

StdP: 14.67

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 34.1	(c) 36.1	(d) 28.1	(e) 22.1	(f) 44.4	(g) 31.6	(h) 26.0	(i) 41.2	(j) 31.8	(k) 53.1	(l) 28.4	(m) 53.5	(n) 3.9	(o) 150	(p) 0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.3	84.3	67.2	60.8	79.8	66.2	59.3	75.6	65.0	30.0	73.6	29.1	71.9	28.3	70.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.0	(b) 21.2	(c) 18.8	(e) 30.7	(f) 83.5	(g) 1.8	(h) 3.1	(i) 29.4	(j) 85.8	(k) 28.4	(l) 87.6	(m) 27.4	(n) 89.4	(o) 26.1	(p) 91.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	62.6	62.0	59.8	55.6	52.9	50.2	48.8	49.8	51.5	54.0	57.3	60.5
	DBStd	5.83	2.82	2.82	3.14	3.57	3.98	4.07	4.00	3.69	3.44	2.97	2.95	2.99
	HDD50	216	0	0	0	3	18	48	72	49	22	3	1	0
	HDD65	3535	84	91	164	282	376	445	503	470	405	340	233	142
	CDD50	2177	392	335	303	171	107	53	34	44	67	128	218	325
	CDD65	21	10	6	2	0	0	0	0	0	0	0	0	3
	CDH74	230	99	56	32	3	0	0	0	0	3	2	7	28
	CDH80	17	8	2	3	0	0	0	0	0	0	0	2	2
Wind	WSAvg	8.1	9.5	8.6	7.6	6.6	6.7	7.7	7.7	7.9	7.8	8.1	9.2	9.5
Precipitation	PrecAvg	42.5	0.6	0.7	0.9	2.5	6.5	8.7	7.8	6.1	3.4	2.4	1.4	1.0
	PrecMax	60.3	3.3	3.8	5.0	9.5	19.7	22.1	19.5	18.0	10.1	7.2	8.2	5.3
	PrecMin	23.3	0.0	0.0	0.0	0.0	1.0	0.1	0.9	0.9	0.3	0.0	0.0	0.0
	PrecStd	10.6	0.8	0.8	1.0	2.0	4.4	4.2	4.0	3.1	1.9	1.9	1.5	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.6	78.5	77.4	73.0	67.1	62.3	61.0	64.6	68.4	70.0	73.7	77.0
		MCWB	64.6	64.1	63.0	60.0	58.9	55.8	53.2	54.1	55.2	59.4	61.1	63.4
	2%	DB	76.7	75.4	73.5	68.4	63.2	59.3	57.6	59.4	63.9	66.5	70.1	73.8
		MCWB	63.7	63.0	61.6	58.6	57.0	54.5	53.0	52.8	54.5	57.2	59.9	62.2
	5%	DB	74.2	73.4	71.3	66.3	61.1	57.6	56.9	57.5	60.9	64.2	68.1	71.8
		MCWB	62.5	61.9	60.5	57.8	56.2	54.1	52.8	52.2	53.5	56.3	58.6	61.2
	10%	DB	72.1	71.5	69.4	64.1	59.3	56.9	55.4	55.8	59.0	62.2	66.2	69.9
		MCWB	61.2	60.8	59.9	57.0	55.6	53.8	52.1	51.7	52.8	55.2	57.4	60.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.0	66.9	65.6	63.3	61.2	58.3	56.6	56.9	58.4	60.8	62.9	65.4
		MCDB	76.7	74.0	73.1	68.1	64.4	60.4	58.3	60.4	64.9	68.4	70.8	74.4
	2%	WB	65.1	64.7	63.4	61.1	59.2	56.6	55.2	55.4	56.4	58.6	60.8	63.5
		MCDB	73.8	72.3	70.7	65.3	61.5	58.2	56.9	57.8	60.9	64.7	67.7	71.7
	5%	WB	63.7	63.3	62.0	59.7	57.9	55.5	54.1	54.3	55.1	57.1	59.6	62.1
		MCDB	71.9	70.9	68.6	63.8	60.1	57.1	55.9	56.6	59.0	62.3	66.3	69.8
	10%	WB	62.4	61.9	60.6	58.3	56.7	54.5	53.1	53.1	54.0	55.8	58.4	60.8
		MCDB	70.1	69.2	66.8	62.4	58.8	56.4	54.9	55.4	57.7	60.6	64.7	68.2
Mean Daily Temperature Range	5% DB	MDBR	19.6	19.1	19.0	17.1	13.9	12.7	13.2	13.6	15.7	16.1	17.9	18.8
		MCDBR	23.4	23.0	23.0	21.2	16.1	13.1	13.7	15.3	20.0	20.4	21.7	22.4
		MCWBR	12.3	12.2	12.8	12.8	10.8	9.6	9.6	10.4	13.0	13.0	12.9	12.4
	5% WB	MCDBR	20.9	19.8	19.7	17.2	12.3	9.2	9.7	11.5	16.3	17.8	19.1	20.0
		MCWBR	11.6	11.1	11.8	11.5	9.1	7.6	7.9	8.8	11.3	11.8	11.8	11.5
Clear-Sky Solar Irradiance	taub	0.323	0.326	0.317	0.322	0.329	0.324	0.316	0.324	0.328	0.323	0.317	0.326	
	taud	2.576	2.566	2.596	2.560	2.538	2.573	2.582	2.552	2.509	2.539	2.579	2.551	
	Ebn at Noon	321	313	304	282	260	252	262	277	295	310	321	321	
	Edh at Noon	33	32	29	26	24	21	22	26	31	33	33	34	
All-Sky Solar Radiation	RadAvg	2624	2271	1802	1201	739	577	668	918	1415	1866	2383	2622	
	RadStd	116	118	86	82	68	76	71	81	96	125	103	128	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+5
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page